

Chapter 9

Linear Programming

Linear programming is a method for finding the best value of a linear objective function subject to linear constraints. In applied settings, these constraints represent limits on resources, time, materials, budget, or other practical conditions. The objective function represents the quantity to be maximized or minimized, such as profit, cost, or output.

This chapter brings together the ideas from the previous chapter and applies them systematically. We identify decision variables, write constraints, graph feasible regions, locate vertices, and evaluate the objective function at those corner points to determine an optimal solution.

9.1 Structure of a Linear Program

Every linear program has three components.

Definition. *Linear program.* A **linear program** consists of:

- (i) **Decision variables:** the quantities being chosen, typically labeled x and y .
- (ii) **Constraints:** linear inequalities that bound the feasible region, expressing limits on resources, time, capacity, or other conditions.
- (iii) **Objective function:** the linear expression $f = ax + by$ to be maximized or minimized.

The solution to a linear program is the point in the feasible region at which the objective function achieves its optimal value. For the two-variable problems in this chapter, that optimal value occurs at a vertex of the feasible region.

The method.

1. Identify the decision variables.
2. Write all constraints as linear inequalities. Include non-negativity constraints.
3. Write the objective function.

4. Find the feasible region and all its vertices.
5. Evaluate the objective function at each vertex.
6. Select the vertex giving the maximum or minimum value as required.
7. Verify that this vertex satisfies all constraints.

Example 9.1.1. (*Setting up a linear program*) A carpenter produces chairs and bookcases. Each chair requires 2 hours of labor and 3 board-feet of wood. Each bookcase requires 4 hours of labor and 6 board-feet of wood. The carpenter has at most 40 hours of labor and at most 72 board-feet of wood available per week. Each chair earns a profit of \$60 and each bookcase earns \$120.

Decision variables: x = chairs per week, y = bookcases per week.

Constraints:

$$2x + 4y \leq 40 \quad (\text{labor hours})$$

$$3x + 6y \leq 72 \quad (\text{board-feet})$$

$$x \geq 0$$

$$y \geq 0$$

Objective: Maximize $\pi = 60x + 120y$.

Note: simplifying the constraints by dividing through: $x + 2y \leq 20$ and $x + 2y \leq 24$. Since every point satisfying $x + 2y \leq 20$ also satisfies $x + 2y \leq 24$, the wood constraint is redundant. The effective constraints are $x + 2y \leq 20$, $x \geq 0$, $y \geq 0$.

Exercises 9.1

9.1.1. For each situation, identify the decision variables, write the constraints, and write the objective function. Do not solve yet.

- (a) A farmer plants corn (x acres) and soybeans (y acres). Each acre of corn requires 3 hours of labor and yields \$200 profit. Each acre of soybeans requires 2 hours of labor and yields \$150 profit. Total labor available: 120 hours. Total land available: 50 acres. Maximize profit.
- (b) A caterer prepares sandwiches (x) and salads (y). Each sandwich uses 2 oz of bread and 3 oz of filling. Each salad uses 1 oz of bread and 4 oz of filling. Available: 80 oz of bread and 150 oz of filling. Each sandwich earns \$4 and each salad earns \$5. Maximize revenue.
- (c) A hospital dietitian plans meals using food A (x servings) and food B (y servings). Each serving of A costs \$2 and provides 3 grams of protein. Each serving of B costs \$3 and provides 5 grams of protein. The meal must provide at least 30 grams of protein. Minimize cost.

9.1.2. A linear program has two decision variables, three resource constraints, and non-negativity constraints. What is the maximum number of vertices the feasible region can have? Explain your reasoning.

9.1.3. Explain in two complete sentences why non-negativity constraints are included in almost every applied linear program.

9.2 Maximization Problems

A maximization problem seeks the largest possible value of the objective function over the feasible region. The optimal solution is the vertex at which the objective is largest.

Example 9.2.1. A furniture maker produces chairs (x) and desks (y). Each chair earns \$40 profit and each desk earns \$70 profit. Constraints:

$$2x + 3y \leq 120, \quad x + 2y \leq 70, \quad x \geq 0, \quad y \geq 0.$$

Objective: Maximize $\pi = 40x + 70y$.

Step 1: Find the vertices.

Intercepts of $2x + 3y = 120$: $(60, 0)$ and $(0, 40)$. Intercepts of $x + 2y = 70$: $(70, 0)$ and $(0, 35)$.

Intersection of the two lines: From $x + 2y = 70$: $x = 70 - 2y$. Substitute: $2(70 - 2y) + 3y = 120 \Rightarrow 140 - 4y + 3y = 120 \Rightarrow y = 20, x = 30$.

Vertices: $(0, 0)$, $(60, 0)$, $(30, 20)$, $(0, 35)$.

Step 2: Check that all vertices are feasible.

$(60, 0)$: $2(60) + 3(0) = 120 \leq 120 \checkmark$, $60 + 2(0) = 60 \leq 70 \checkmark$.

$(0, 35)$: $2(0) + 3(35) = 105 \leq 120 \checkmark$, $0 + 2(35) = 70 \leq 70 \checkmark$.

$(30, 20)$: $2(30) + 3(20) = 120 \leq 120 \checkmark$, $30 + 2(20) = 70 \leq 70 \checkmark$.

All vertices are feasible.

Step 3: Evaluate the objective.

Vertex (x, y)	Profit $\pi = 40x + 70y$
$(0, 0)$	0
$(60, 0)$	2,400
$(30, 20)$	$1,200 + 1,400 = 2,600$
$(0, 35)$	2,450 $\leftarrow$ maximum?

Wait: $40(30) + 70(20) = 1,200 + 1,400 = 2,600$. And $40(0) + 70(35) = 2,450$. The maximum is $\pi = \$2,600$ at $(30, 20)$.

The furniture maker should produce 30 chairs and 20 desks per week to maximize profit.

Example 9.2.2. (*Recognizing a redundant constraint*) A print shop produces posters (x) and banners (y). Constraints:

$$x + 2y \leq 60, \quad x + y \leq 40, \quad 2x + y \leq 70, \quad x \geq 0, \quad y \geq 0.$$

Objective: Maximize $\pi = 15x + 35y$.

Find intersections to locate vertices. $x + 2y = 60$ and $x + y = 40$: subtract to get $y = 20$, $x = 20$. Vertex $(20, 20)$.

Check $(20, 20)$ against $2x + y \leq 70$: $2(20) + 20 = 60 \leq 70$. ✓

$x + y = 40$ and $2x + y = 70$: subtract to get $x = 30$, $y = 10$. Candidate vertex $(30, 10)$.

Check $(30, 10)$ against $x + 2y \leq 60$: $30 + 20 = 50 \leq 60$. ✓

Vertices of the feasible region: $(0, 0)$, $(35, 0)$, $(30, 10)$, $(20, 20)$, $(0, 30)$.

Note: $(35, 0)$ comes from $2x \leq 70$, i.e., $x \leq 35$. Check against $x + y \leq 40$: $35 + 0 = 35 \leq 40$. ✓. And $x + 2y \leq 60$: $35 \leq 60$. ✓.

Vertex	$\pi = 15x + 35y$
$(0, 0)$	0
$(35, 0)$	525
$(30, 10)$	$450 + 350 = 800$
$(20, 20)$	$300 + 700 = 1,000 \leftarrow \text{maximum}$
$(0, 30)$	1,050

Wait: $(0, 30)$: $15(0) + 35(30) = 1,050$. Check feasibility: $0 + 2(30) = 60 \leq 60$ ✓, $0 + 30 = 30 \leq 40$ ✓, $0 + 30 = 30 \leq 70$ ✓.

The maximum is $\pi = \$1,050$ at $(0, 30)$. The shop should produce 0 posters and 30 banners.

Exercises 9.2

9.2.1. Solve each maximization problem completely. Show all steps: vertices, feasibility checks, and objective evaluation.

(a) Maximize $\pi = 5x + 8y$ subject to: $x + 2y \leq 16$, $3x + 2y \leq 24$, $x \geq 0$, $y \geq 0$.

(b) Maximize $\pi = 10x + 6y$ subject to: $x + y \leq 12$, $2x + y \leq 18$, $x \geq 0$, $y \geq 0$.

(c) Maximize $R = 20x + 30y$ subject to: $2x + 3y \leq 60$, $x + y \leq 24$, $x \geq 0$, $y \geq 0$.

9.2.2. A farm grows tomatoes (x crates) and peppers (y crates) subject to:

$$x + 2y \leq 80, \quad x + y \leq 50, \quad x \geq 0, \quad y \geq 0.$$

Each crate of tomatoes earns \$25 and each crate of peppers earns \$40.

- (a) Write the objective function.
- (b) Find all vertices of the feasible region.
- (c) Determine the production plan that maximizes revenue.

9.2.3. A linear program has objective $\pi = 3x + 3y$ and feasible region with vertices $(0, 0)$, $(4, 0)$, $(3, 2)$, $(0, 5)$.

- (a) Evaluate π at all four vertices.
- (b) Describe what happens when the objective has equal coefficients for x and y . Why might the maximum not be unique?

9.3 Minimization Problems

A minimization problem seeks the smallest possible value of the objective function. The method is identical to maximization: find the vertices, evaluate the objective, and select the minimum. Minimization problems often arise when the modeler wants to reduce cost, time, or resource use while satisfying minimum requirements.

Minimization problems frequently have constraints of the form $ax + by \geq c$ (the region must satisfy a minimum requirement) rather than $ax + by \leq c$ (an upper limit). These constraints point outward, creating an unbounded feasible region. The minimum of a linear objective over an unbounded region still occurs at a vertex, provided the objective is bounded below on the region.

Example 9.3.1. A hospital dietitian plans daily meals from two food sources: A and B. Each unit of A costs \$2 and contains 3 g protein and 1 g fiber. Each unit of B costs \$3 and contains 1 g protein and 2 g fiber. Requirements: at least 12 g protein and at least 8 g fiber.

Decision variables: x = units of A, y = units of B.

Constraints:

$$3x + y \geq 12, \quad x + 2y \geq 8, \quad x \geq 0, \quad y \geq 0.$$

Objective: Minimize $C = 2x + 3y$.

Finding the vertices.

Intercepts of $3x + y = 12$: $(4, 0)$ and $(0, 12)$. Intercepts of $x + 2y = 8$: $(8, 0)$ and $(0, 4)$.

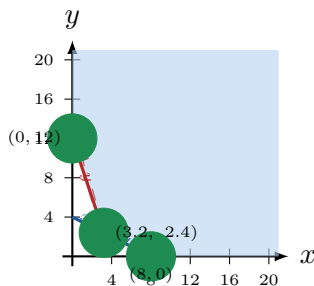
Intersection of $3x + y = 12$ and $x + 2y = 8$: From the first: $y = 12 - 3x$. Substitute: $x + 2(12 - 3x) = 8 \Rightarrow x + 24 - 6x = 8 \Rightarrow -5x = -16 \Rightarrow x = 3.2$, $y = 12 - 9.6 = 2.4$.

Check: does $(8, 0)$ satisfy $3x + y \geq 12$? $3(8) + 0 = 24 \geq 12$. ✓

Does $(4, 0)$ satisfy $x + 2y \geq 8$? $4 + 0 = 4 \not\geq 8$. $\times$

So $(4, 0)$ is not feasible.

The vertices of the feasible region are: $(8, 0)$, $(3.2, 2.4)$, $(0, 12)$.



Vertex	$C = 2x + 3y$
$(8, 0)$	$16 + 0 = 16$
$(3.2, 2.4)$	$6.4 + 7.2 = 13.6 \leftarrow \text{minimum}$
$(0, 12)$	$0 + 36 = 36$

The minimum cost of \$13.60 is achieved at $x = 3.2$ units of A and $y = 2.4$ units of B.

Example 9.3.2. (*Minimization with an upper-bounded region*) A school orders buses for a field trip using large buses (x) and small buses (y). Large buses hold 50 students and cost \$400. Small buses hold 30 students and cost \$250. At least 300 students must be transported. At most 8 buses total are available.

Constraints:

$$50x + 30y \geq 300, \quad x + y \leq 8, \quad x \geq 0, \quad y \geq 0.$$

Objective: Minimize $C = 400x + 250y$.

Simplify the first constraint by dividing by 10: $5x + 3y \geq 30$.

Find the vertices. Intercepts of $5x + 3y = 30$: $(6, 0)$ and $(0, 10)$.

Does $(6, 0)$ satisfy $x + y \leq 8$? $6 \leq 8$. $\checkmark$

Does $(0, 10)$ satisfy $x + y \leq 8$? $10 \not\leq 8$. $\times$

Intersection of $5x + 3y = 30$ and $x + y = 8$: From the second: $x = 8 - y$. $5(8 - y) + 3y = 30 \Rightarrow 40 - 5y + 3y = 30 \Rightarrow y = 5$, $x = 3$. Vertex $(3, 5)$.

Intersection of $x + y = 8$ and x -axis: $(8, 0)$. Check: $5(8) + 3(0) = 40 \geq 30$. $\checkmark$

Vertices: $(6, 0)$, $(8, 0)$, $(3, 5)$.

Vertex	$C = 400x + 250y$	
(6, 0)	2,400	$\leftarrow$ minimum
(8, 0)	3,200	
(3, 5)	$1,200 + 1,250 = 2,450$	

The minimum cost of \$2,400 uses 6 large buses and 0 small buses.

Exercises 9.3

9.3.1. Solve each minimization problem completely.

- (a) Minimize $C = 3x + 5y$ subject to: $x + y \geq 8$, $2x + y \geq 10$, $x \geq 0$, $y \geq 0$.
- (b) Minimize $C = 4x + 2y$ subject to: $2x + y \geq 12$, $x + 3y \geq 12$, $x \geq 0$, $y \geq 0$.
- (c) Minimize $C = 6x + 9y$ subject to: $x + 2y \geq 10$, $3x + y \geq 12$, $x \geq 0$, $y \geq 0$.

9.3.2. A nutritionist plans daily servings of two supplements: X and Y. Supplement X costs \$1.50 per serving and provides 4 mg of iron and 2 mg of zinc. Supplement Y costs \$2.00 per serving and provides 2 mg of iron and 5 mg of zinc. Daily requirements: at least 16 mg iron and at least 15 mg zinc.

- (a) Write the constraints and objective function.
- (b) Find the vertices of the feasible region.
- (c) Determine the supplement plan that minimizes daily cost.

9.3.3. A city uses large trucks (x) and small trucks (y) for waste collection. Large trucks collect 8 tons per trip and cost \$300. Small trucks collect 3 tons per trip and cost \$150. At least 24 tons must be collected. At most 6 trucks are available.

- (a) Write the linear program.
- (b) Find the vertices of the feasible region.
- (c) Minimize total cost.

9.4 Applications and Modeling

A linear program begins with a real situation, not with equations. The modeler's first task is to read the situation carefully and translate it into the language of variables, constraints,

and an objective. This section develops that translation skill through progressively more complex situations.

Guidelines for setting up a linear program.

1. Read the problem and identify what is being decided. These are the decision variables.
2. Identify the limited resources or requirements. Each one produces a constraint.
3. Identify the quantity to be optimized. This is the objective function.
4. Check units: the coefficients in each constraint must be in the same units.

Example 9.4.1. (*Production planning*) A toy company makes two products: action figures (x) and toy cars (y). Each action figure requires 3 minutes of machine time and 2 minutes of assembly time. Each toy car requires 1 minute of machine time and 4 minutes of assembly time. The machine is available for at most 120 minutes per day. Assembly workers are available for at most 160 minutes per day. Each action figure earns \$8 profit and each toy car earns \$5 profit.

Decision variables: x = action figures per day, y = toy cars per day.

Constraints:

$$3x + y \leq 120, \quad 2x + 4y \leq 160, \quad x \geq 0, \quad y \geq 0.$$

Simplify the second: $x + 2y \leq 80$.

Objective: Maximize $\pi = 8x + 5y$.

Vertices. $3x + y = 120$ intercepts: $(40, 0)$ and $(0, 120)$. $x + 2y = 80$ intercepts: $(80, 0)$ and $(0, 40)$.

Intersection: From $x + 2y = 80$: $x = 80 - 2y$. $3(80 - 2y) + y = 120 \Rightarrow 240 - 6y + y = 120 \Rightarrow y = 24, x = 32$.

Check $(0, 40)$ against $3x + y \leq 120$: $0 + 40 = 40 \leq 120$. ✓

Check $(40, 0)$ against $x + 2y \leq 80$: $40 \leq 80$. ✓

Vertices: $(0, 0)$, $(40, 0)$, $(32, 24)$, $(0, 40)$.

Vertex	$\pi = 8x + 5y$
$(0, 0)$	0
$(40, 0)$	320
$(32, 24)$	$256 + 120 = 376 \leftarrow$ maximum
$(0, 40)$	200

The company should produce 32 action figures and 24 toy cars per day for maximum profit of \$376.

Example 9.4.2. (*Resource allocation with minimum requirements*) A municipality allocates funding between road repairs (x thousand dollars) and park maintenance (y thousand dollars). Total budget: at most \$500 thousand. Policy requires at least \$100 thousand for roads and at least \$80 thousand for parks. Each thousand dollars in roads generates 3 units of public satisfaction; each thousand in parks generates 4 units.

Constraints:

$$x + y \leq 500, \quad x \geq 100, \quad y \geq 80.$$

Objective: Maximize $S = 3x + 4y$.

The constraint $x \geq 100$ is a lower bound on x . Similarly $y \geq 80$. These shift the feasible region away from the origin.

Vertices:

- $x = 100, y = 80$: $(100, 80)$. Check $x + y \leq 500$: $180 \leq 500$. ✓
- $x = 100, y = 400$ (from $x + y = 500$ with $x = 100$).
- $x = 420, y = 80$ (from $x + y = 500$ with $y = 80$).

Vertex	$S = 3x + 4y$
$(100, 80)$	$300 + 320 = 620$
$(100, 400)$	$300 + 1,600 = 1,900$
$(420, 80)$	$1,260 + 320 = 1,580$

Maximum satisfaction of 1,900 units: allocate \$100 thousand to roads and \$400 thousand to parks. The parks generate more satisfaction per dollar, so the optimal solution devotes as much funding as possible to parks while meeting the minimum road requirement.

Exercises 9.4

9.4.1. A print shop produces two types of brochures: standard (x) and premium (y). Each standard brochure uses 2 minutes of press time and 1 minute of finishing time. Each premium brochure uses 3 minutes of press time and 4 minutes of finishing time. Press time available: 120 minutes. Finishing time available: 100 minutes. Standard earns \$1.50 profit; premium earns \$4 profit.

- Write the linear program.
- Find all vertices.

(c) Determine the production plan that maximizes profit.

9.4.2. A warehouse stores two products: A and B. Product A occupies 4 cubic feet and weighs 10 pounds. Product B occupies 6 cubic feet and weighs 5 pounds. The warehouse has 240 cubic feet of space and can support at most 500 pounds. Product A generates \$12 of value per unit and Product B generates \$9.

(a) Write the linear program.

(b) Find all vertices of the feasible region.

(c) Maximize total value.

9.4.3. A city council divides a \$600,000 grant between job training programs (x thousand dollars) and housing subsidies (y thousand dollars). Policy requires at least \$150,000 for each program. Job training generates 2 employment outcomes per thousand dollars. Housing subsidies generate 1.5 family placements per thousand dollars. Both outcomes are valued equally by the council.

(a) Write the linear program, using $f = 2x + 1.5y$ as the objective.

(b) Find the vertices.

(c) Solve. Interpret the result in one sentence.

9.5 The Modeler and the Program

A linear program encodes a situation through the choice of decision variables, constraints, and objective function. Changing the objective function changes the question being asked and, consequently, the optimal vertex. The feasible region remains fixed; the modeler's question determines the answer.

Modeler	Objective	Action
Business owner	Maximize profit	Find max vertex
Operations manager	Minimize cost	Find min vertex
Logistics planner	Minimize time	Find min vertex
Revenue analyst	Maximize revenue	Find max vertex
Policy maker	Maximize social benefit	Find max vertex

The same feasible region can produce different optimal solutions for different modelers. Different objective functions emphasize different goals, so it is important to identify clearly what is being optimized before solving the problem. In applications, choosing the correct

objective function is part of the modeling process. interval depending on the question. Here, the same feasible region yields different vertices depending on the objective.

Example 9.5.1. A workshop has the feasible region with vertices $(0, 0)$, $(40, 0)$, $(30, 20)$, $(0, 35)$ from the furniture maker in Example 9.2.1.

The same workshop, three modelers.

Modeler 1 (profit): Maximize $\pi = 40x + 70y$. Maximum at $(30, 20)$: $\pi = \$2,600$.

Modeler 2 (revenue): Maximize $R = 60x + 100y$.

$$(0, 0): 0 \qquad (40, 0): 2,400$$

$$(30, 20): 3,800 \quad (0, 35): 3,500$$

Maximum at $(30, 20)$: $R = \$3,800$.

Modeler 3 (material cost): Minimize $C = 8x + 12y$.

$$(0, 0): 0 \qquad (40, 0): 320$$

$$(30, 20): 480 \quad (0, 35): 420$$

Minimum at $(0, 0)$: $C = \$0$. As in Chapter 8, the minimum-at-origin result reflects zero production; a practical floor on production would add a lower-bound constraint.

Modelers 1 and 2 both find the maximum at $(30, 20)$ for this particular region and these particular objectives. This coincidence does not hold in general.

Exercises 9.5

9.5.1. Using the feasible region with vertices $(0, 0)$, $(8, 0)$, $(5, 4)$, $(0, 6)$:

(a) Maximize $\pi = 10x + 12y$.

(b) Maximize $R = 15x + 8y$.

(c) Minimize $C = 4x + 6y$ (ignoring the zero-production vertex).

(d) Do all three objectives lead to the same optimal vertex? What does the difference tell you about these three questions?

9.5.2. The toy company from Example 9.4.1 considers a new pricing scenario. Action figures now earn \$5 profit and toy cars earn \$8 profit. The constraints remain the same.

(a) Write the new objective function.

(b) Evaluate at all vertices.

- (c) Compare the optimal solution with the original (Example 9.4.1). What changed and why?

Chapter 9 Review

1. Solve each linear program. Show all steps.
 - (a) Maximize $\pi = 6x + 9y$ subject to: $2x + 3y \leq 24$, $x + y \leq 10$, $x \geq 0$, $y \geq 0$.
 - (b) Minimize $C = 5x + 4y$ subject to: $x + 2y \geq 12$, $2x + y \geq 10$, $x \geq 0$, $y \geq 0$.
2. A bakery produces two items: loaves (x) and pies (y). Each loaf needs 1 hour of baking time and uses 2 cups of flour. Each pie needs 2 hours of baking time and uses 3 cups of flour. Baking time available: 24 hours. Flour available: 36 cups. Profit: \$5 per loaf, \$9 per pie.
 - (a) Write the linear program.
 - (b) Find all vertices of the feasible region.
 - (c) Maximize profit.
 - (d) How much of each resource is used at the optimal solution? Is either resource fully used?
3. A shipping company has two types of containers: standard (x) and refrigerated (y). Standard containers weigh 2 tons and generate \$300 revenue. Refrigerated containers weigh 5 tons and generate \$700 revenue. The ship can carry at most 100 tons. The company must ship at least 10 refrigerated containers per voyage.
 - (a) Write the linear program.
 - (b) Find the vertices.
 - (c) Maximize revenue.
4. A city park service assigns staff to two tasks: mowing (x hours) and planting (y hours). Each hour of mowing costs \$20 and covers 500 square feet. Each hour of planting costs \$35 and adds 50 plants. The park needs at least 4,000 square feet mowed and at least 200 plants added. The total labor budget is at most \$600.
 - (a) Write the constraints and objective (minimize cost).
 - (b) Find the vertices.
 - (c) Solve.
5. A linear program has feasible region with vertices $(2, 8)$, $(6, 4)$, $(10, 0)$.

- (a) Maximize $f = 3x + 5y$.
 - (b) For what values of a does $f = ax + 5y$ achieve its maximum at $(2, 8)$ rather than $(6, 4)$? Set up and solve the inequality $f(2, 8) > f(6, 4)$.
6. Explain in three complete sentences how linear programming brings together the tools of Chapters 7 and 8. Identify the specific role that each chapter's contribution plays in the method.
7. A student sets up a linear program and finds that the objective function achieves the same value at two adjacent vertices.
- (a) What does this tell you about the optimal solution?
 - (b) Give an example of an objective function and two vertices where this would occur.